# 综合练习

## 人员

王承周、王毅博、阮文璋 到课

## 上周作业检查

上上周作业链接: https://vjudge.net/contest/801929

开始时间 : 2026-04-05 08:30 CST

2026-0405 六队上课 (综合练习)

结束时

已开始 : 14:05:00:17

进行中

概览 题目 提交状态 排名 (14:05:00:11) 2 条评论

| Rank | Team | Score | Penalty | A 10 / 13 | B 9 / 9 | C 5 / 7 | D 6 / 9 | E 7 / 11 |
|---|---|---|---|---|---|---|---|---|
| 1 | yyycj | 5 | 40888 | 1:13:00 | 7:01:34:54 | 7:04:46:18 | 7:01:29:19 | 7:00:25:16 |
| 2 | chenxinmiao (陈欣妙) | 5 | 43426 | 2:14:02:16 (-3) | 5:08:37:00 | 7:13:17:32 (-1) | 7:12:42:38 | 7:01:27:00 (-1) |
| 3 | r111 | 5 | 59734 | 5:11:08:35 | 9:12:58:00 | 9:14:48:00 | 9:13:48:00 | 7:06:52:00 |
| 4 | lyc12345bc (刘奕辰) | 5 | 69665 | 13:14:33:23 | 7:01:50:30 | 13:14:21:00 (-1) | 7:01:29:53 | 7:00:30:29 |
| 5 | lijinshu | 5 | 78013 | 6:12:12:31 | 13:13:09:00 | 13:14:06:08 | 13:12:12:52 | 7:00:33:11 |
| 6 | wcz123bc | 4 | 57397 | 6:05:44:35 | 13:03:58:56 | | 13:04:23:54 | 7:06:30:19 |
| 7 | junyan007008 | 3 | 20461 | 2:36:00 | 7:01:42:43 | | (-3) | 7:00:42:55 |
| 8 | wyb1111 (王毅博) | 2 | 20869 | 7:04:55:00 | 7:06:54:00 | | | |
| 9 | xuruiqian | 2 | 20883 | 7:05:29:00 | 7:06:34:00 | | | |
| 10 | chujinxuan (褚锦轩) | 1 | 10105 | 7:00:25:08 | | | | (-3) |

上周作业链接: https://vjudge.net/contest/803886

## 本周作业

https://vjudge.net/contest/805933 (课上讲了上周的 A ~ E 题, 课后作业是本周的 A ~ E 题)

## 课堂表现

这周的题目整体难度要小一些, 同学们课上听讲做题整体都比较好, 课下要记得把课上没做完的题补完。

## 课堂内容

**CF30C Shooting Gallery**

按照出现时间排序, 然后跑 n^2 的最长上升子序列即可

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int maxn = 1000 + 5;
struct node {
  int x, y, t;
  double p;
  bool operator < (const node& p) const { return t < p.t; }
} w[maxn];
double f[maxn];

bool check(node a, node b) {
  int x1 = a.x, y1 = a.y, t1 = a.t, x2 = b.x, y2 = b.y, t2 = b.t;
  int dx = x1 - x2, dy = y1 - y2, dt = t1 - t2;
  return (LL)dx*dx + (LL)dy*dy <= (LL)dt*dt;
}

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) {
    cin >> w[i].x >> w[i].y >> w[i].t >> w[i].p;
  }
  sort(w+1, w+n+1);

  double res = 0;
  for (int i = 1; i <= n; ++i) {
    for (int j = 1; j <= i-1; ++j) {
      if (check(w[j], w[i])) f[i] = max(f[i], f[j]);
    }
    f[i] += w[i].p;
    res = max(res, f[i]);
  }

  printf("%.8f\n", res);
  return 0;
}
```

**CF1884B Haunted House**

求每次把 0 往后移的代价是多少

```
#include <bits/stdc++.h>

using namespace std;
```

```
typedef long long LL;
const int maxn = 1e5 + 5;
char s[maxn];

void solve() {
  int n; cin >> n >> (s+1);

  LL res = 0;
  for (int i = n, j = n; i >= 1; --i) {
    while (j>=1 && s[j]!='0') --j;
    if (j>=1) {
      res += i-j;
      cout << res << " ";
      --j;
    }
    else {
      for (int k = i; k >= 1; --k) cout << -1 << " ";
      break;
    }
  }
  cout << endl;
}

int main()
{
  int T; cin >> T;
  while (T -- ) solve();
  return 0;
}
```

**CF1903D1 Maximum And Queries (easy version)**

贪心, 按照每个数的二进制进行考虑, 从最高位开始, 看能否把每一位都变为 1, 能变则一定变

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int maxn = 1e5 + 5;
LL w[maxn], a[maxn];

LL solve(int n, LL k) {
  for (int i = 1; i <= n; ++i) a[i] = w[i];

  LL res = 0;
  for (int i = 61; i >= 0; --i) {
    LL sum = 0;
    for (int j = 1; j <= n; ++j) {
      if ((a[j]>>i)&1) continue;
      sum += (1LL<<i) - a[j]%(1LL<<i);
```

```
      if (sum > k) break;
    }
    if (sum > k) continue;

    k -= sum; res += (1LL<<i);
    for (int j = 1; j <= n; ++j) {
      if ((a[j]>>i)&1) continue;
      a[j] += (1LL<<i) - a[j]%(1LL<<i);
    }
  }
  return res;
}

int main()
{
  int n, T; cin >> n >> T;
  for (int i = 1; i <= n; ++i) cin >> w[i];

  while (T -- ) {
    LL k; cin >> k;
//    cout << "--------------------- ";
    cout << solve(n, k) << endl;
  }
  return 0;
}
```

**CF448C Painting Fence**

递归考虑处理 [l,r] 最少需要几步

两种方案:

1. 全竖着刷
2. 先横着刷最小的, 然后根据最小的分两边继续递归

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 5000 + 5;
int w[maxn];

int dfs(int l, int r) {  // 处理 [l,r] 最少需要几步
  if (l > r) return 0;

  int minn = 1e9+100;
  for (int i = l; i <= r; ++i) minn = min(minn, w[i]);
  for (int i = l; i <= r; ++i) w[i] -= minn;
  int pos = -1;
  for (int i = l; i <= r; ++i) {
```

```
    if (!w[i]) { pos = i; break; }
  }

  return min(minn+dfs(l,pos-1)+dfs(pos+1,r), r-l+1);
}

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i];
  cout << dfs(1, n) << endl;
  return 0;
}
```

**CF1900C Anji's Binary Tree**

dfs(u): 从 u 到叶子最少需要改几个

1. 如果 u 是叶子, 返回 0
2. 如果 u 只有右儿子, 返回往右走的解
3. 如果 u 只有左儿子, 返回往左走的解
4. 如果 u 两个儿子都有, 返回往左往右解的最小值

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 3e5 + 5;
char s[maxn];
int tr[maxn][2];

int dfs(int u) {
  if (tr[u][0]==0 && tr[u][1]==0) return 0;
  if (tr[u][0] == 0) return (s[u]!='R') + dfs(tr[u][1]);
  if (tr[u][1] == 0) return (s[u]!='L') + dfs(tr[u][0]);
  return min((s[u]!='R') + dfs(tr[u][1]), (s[u]!='L') + dfs(tr[u][0]));
}

void solve() {
  int n; cin >> n;

  cin >> (s+1);
  for (int i = 1; i <= n; ++i) cin >> tr[i][0] >> tr[i][1];

//  cout << "---------------------------- ";
  cout << dfs(1) << endl;
}

int main()
```

```
{
  int T; cin >> T;
  while (T -- ) solve();
  return 0;
}
```